

Research and Collaboration Program Overview

Theory, Applications, and Industrial Outreach.

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- Game theory and economic modeling
 - * Configurations of a game
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 - * Variational analysis of equilibrium points

Optimization: Theory and Modeling

- Optimization modeling for Industrial and energy applications
(Process- & data-driven || Digital twins)
 - * Energy economics
 - * Energy and environmental management
 - * Production and planning
 - * Supply chain and logistics optimization
 - * Experimental designs
 - * etc.

Optimization as a service

What do we do?

When you are unsure of your choices or have a difficulty reaching your goal
— we help designing your *best solution*.

How does it work?

- **Capacity building:** We give a course in optimization, tailored for your need.
- **Research project:** We work closely with you to gain insights, create an *explainable* optimization model, and solve your problem sustainably.

Our strength.

We are mathematicians — we are not bounded to off-the-shelf methods and softwares.
We are ready to craft new models and develop new tools that suit your needs.

Fact.

We are not an expert in *black-box* AI modeling.
We value *transparency* in our modeling processes.

Optimization.

Among the “feasible choices” ,
which one is “the best” ?

$\Leftrightarrow$

$$\left\{ \begin{array}{ll} \max / \min & j(u) \\ \text{s.t.} & u \in U^{\text{ad}}. \end{array} \right.$$

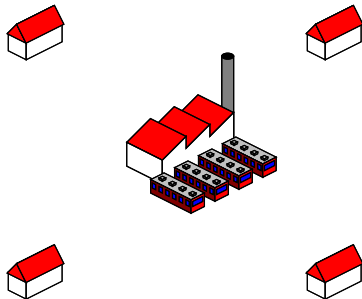
A toy problem

- There are demands, spatially and equally distributed.
- There are four delivery vans.
- **Decision:** A location to construct a new supply warehouse to deliver products.



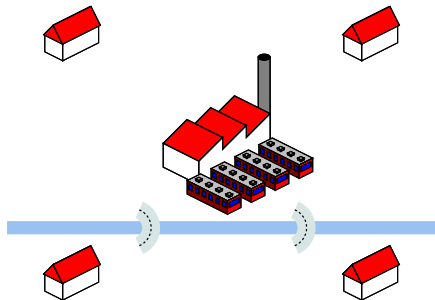
A toy problem

- **Objective:** To minimize the delivery time.
- **Constraint:** All demand points are covered.



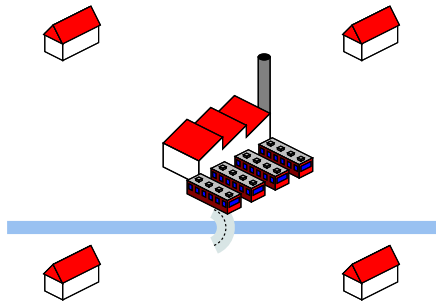
A toy problem

- **Objective:** To minimize the delivery time.
- **Constraints:** All demand points are covered.



A toy problem

- **Objective:** To minimize the delivery time.
- **Constraints:**
 - * All demand points are covered.
 - * Limited budget.



Multi-agent optimization.

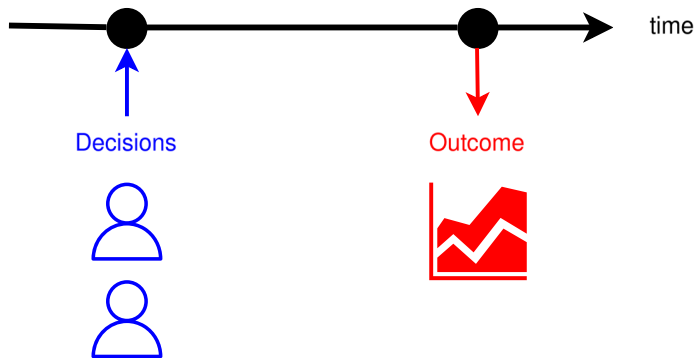
Agent = Decision maker

There are many possible **configurations** of agents.

There is nothing that is **best** to all the agents.
Instead of an **optimal** point, we focus on an **equilibrium**.

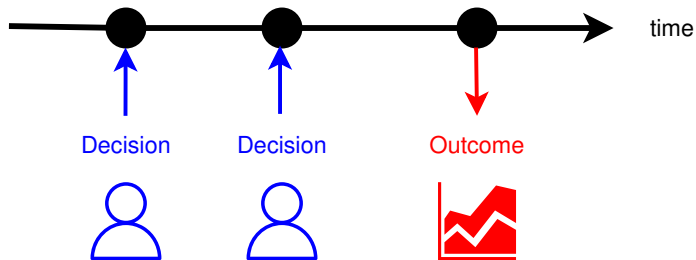
Multi-agent optimization configurations

Nash equilibrium: Agents make independent decisions simultaneously.



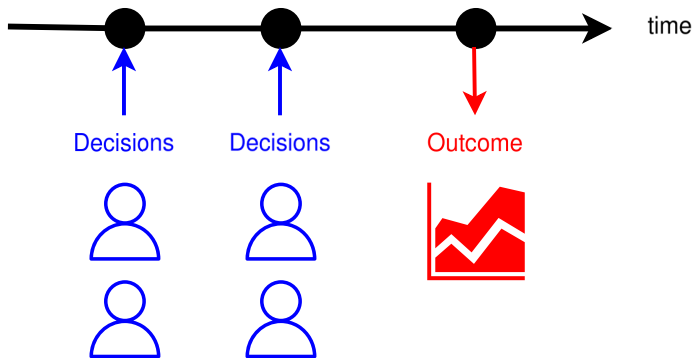
Multi-agent optimization configurations

Stackelberg equilibrium: Agents make a sequential decision.



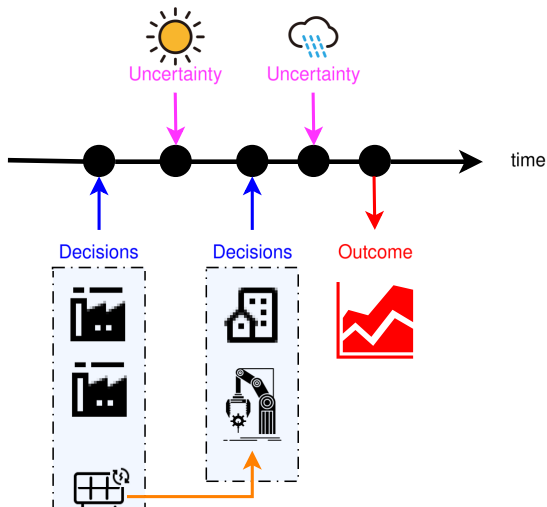
Multi-agent optimization configurations

nash-stackelberg equilibrium: group of agents make decisions simultaneously and sequentially.



Multi-agent optimization configurations

Another complicate configuration (No specific names)



Leader.	1	
Follower.	$f \in F$	
Action sets.	U^a	$a \in A = \{1\} \cup F$
Objectives.	$j^a : U^a \times U^{-a} \rightarrow \mathbb{R}$	$U^{-a} = \prod_{b \neq a} U^b$
Constraints.	$U_{ad}^a : U^{-a} \rightarrow 2^{U^a}$	

Find $(u^a)_{a \in A}$ in which for each $a \in A$, u^a solves

$$\begin{cases} \min_{\hat{u}^a} & j^a(\hat{u}^a, u^{-a}) \\ \text{s.t.} & \hat{u}^a \in U_{\text{ad}}^a(u^{-a}). \end{cases}$$

$$A = \{1\} \cup F.$$

Optimistic Stackelberg game formulation:

$$\begin{cases} \min_{u^1, u^F} & j^1(u^1, u^F) \\ \text{s.t.} & u^1 \in U_{\text{ad}}^1 \\ & u^F \in \text{NE}(u^1), \end{cases}$$

where $u^F = (u^f)_{f \in F} \in \text{NE}(u^1)$ if for all $f \in F$, u^f solves

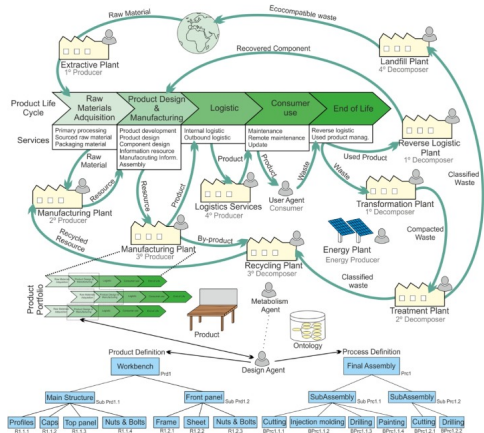
$$\begin{cases} \min_{\hat{u}^f} & j^f(u^1, \hat{u}^f, u^{F \setminus \{f\}}) \\ \text{s.t.} & \hat{u}^f \in U_{\text{ad}}^f(u^1, u^{F \setminus \{f\}}). \end{cases}$$

Theoretical aspects.

- Existence and stability of solutions.
- Reformulation techniques.
- Decision and information structures.
- Witsenhausen models.
- Numerical techniques.

Eco-Industrial Park (EIP) water network: ENSIACET

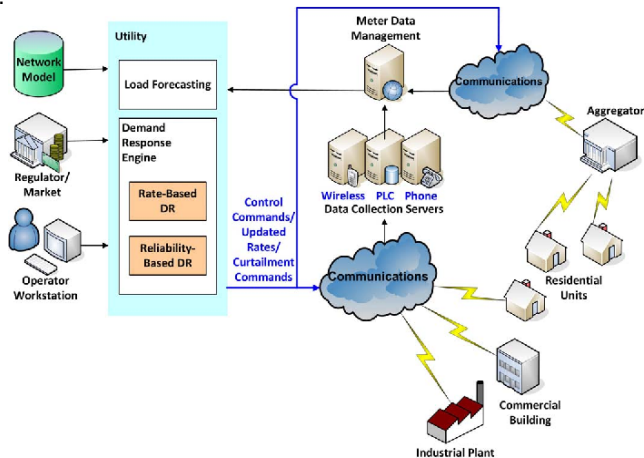
Design circulatory scheme in an industrial park and manage flows of resources.



- Leader: EIP authority
 - ◇ Objective: Minimize clean resources and contamination (Ecological).
 - ◇ Constraints: Resource limitation, Site limitation, etc.
- Followers: Factories
 - ◇ Objective: Minimize payment (Economic).
 - ◇ Constraints: Circulatory limitation, Contamination level, Inventory, etc.

Leader-Follower modeling of demand response: EDF

Information and knowledge structure modeling in the process of demand manipulation.



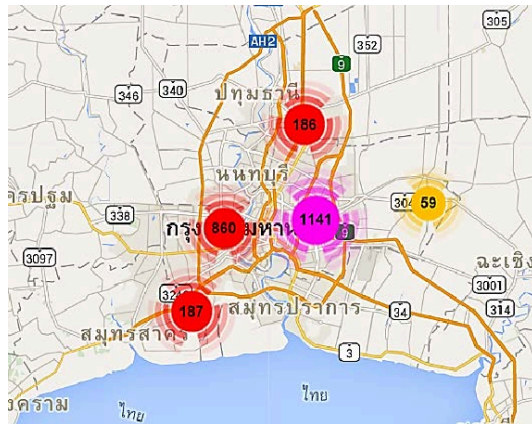
A new thought for Storage management: EDF

A study of how storage's information affects optimal decisions the energy producers.



Optimal surveyor posts: Dhipaya insurance

Place surveyors into posts across Bangkok area so that it covers the dangerous zones with fast-reaching constraints.

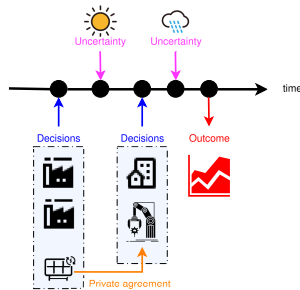


Efficient train control: WCE

Minimize the fuel consumption respecting the schedule.



Stochastic and bilevel optimization techniques in energy management: MEA



- Stochastic optimization course
- Demand response from the point of view of an aggregator.
- Real-time pricing model policy perspectives.
- Smart EV charging program for grid stabilization.



Thank you. :)